

Online Conversion Rate Prediction via Neural Satellite Networks in Delayed Feedback Advertising

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ABSTRACT

The delayed feedback is becoming one of the main obstacles in online advertising due to the pervasive deployment of the cost-per-conversion display strategy requesting a real-time conversion rate (CVR) prediction. It makes the observed data contain a large number of fake negatives that temporarily have no feedback but will convert later. Training on such biased data distribution would severely harm the performance of models. Prevailing approaches wait for a set period of time to see if samples convert before training on them, but solutions to guaranteeing data freshness remain under-explored by current research. In this work, we propose Delayed Feedback modeling via neural Satellite Networks (DFSN for short) for online CVR prediction. It tackles the issue of data freshness to permit adaptive waiting windows. We first assign a long waiting window for our main model to cover most of conversions and greatly reduce fake negatives. Meanwhile, two kinds of satellite models are devised to learn from the latest data, and online transfer learning techniques are utilized to sufficiently exploit their knowledge. With information from satellites, our main model can deal with the issue of data freshness, achieving better performance than previous methods. Extensive experiments on two real-world advertising datasets demonstrate the superiority of our model.

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CCS CONCEPTS

• Information systems → Computational advertising.

KEYWORDS

CVR modeling, Delayed Feedback, Online Advertising

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1 INTRODUCTION

Online advertising is imperative in intelligent e-commerce by enabling advertisers and platforms to more effectively target potential buyers [1, 10, 14, 22, 34]. In an online advertising system, conversion rate (CVR) prediction is at its heart due to the pervasive deployment of the cost-per-conversion (CPA) display strategy that allows advertisers to bid on preset conversion activities, e.g., purchase or download APP, etc [2, 8, 9, 12]. Meanwhile, online advertising systems usually demand a streaming learning paradigm that can be continuously updated with the most recent data to provide a real-time conversion rate prediction [35].

As a consequence, the *delayed feedback* [19, 31] problem is becoming one of the main obstacles. That is, the system may receive the conversion feedback with a delay ranging from a few minutes to several days after an ad is shown to or clicked by a user [13]. For example, Figure 1 exhibits the feedback proportion w.r.t the delayed time ratio on two widely used benchmark datasets. We can observe a clear existence of delayed feedback on these two datasets. The

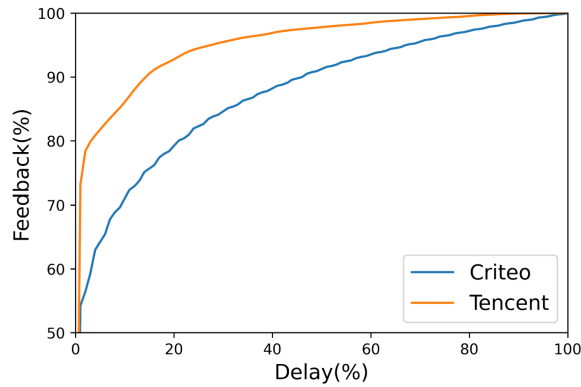


Figure 1: Cumulative conversion distribution of datasets, where Delay(%) denotes the delayed time compared with the longest delayed time of datasets.

longest delayed time in Criteo dataset could reach 30 days, which is much longer than that of Tencent dataset (c.f. Table 1).

The delayed feedback problem makes the observed data contain a large number of fake negatives [17] that temporarily have no feedback but will convert later. Training on such biased data distribution would severely harm the performance of models.

Various solutions were proposed to tackle the delayed feedback problem in existing studies. Early approaches use jointly-trained models to estimate the delay time for better CVR prediction [4, 33], but they are still trained on biased observation data and miss real delayed feedback. Besides, the waiting window is a popular technique for online delayed feedback; models wait a preset period of time to see if samples convert before training on them [5, 11]. The trade-off between label accuracy and data freshness depends on the length of the waiting window. Models with a long waiting window are more likely to receive the actual label of samples and produce fewer fake negatives, while training models on stale data would seriously impair their performance [30]. The newest sample can be obtained by models with a small waiting window, but they would mistakenly classify the samples with a delayed conversion as negative and neglect massive valuable positive samples.

For streaming learning, previous online models mostly choose no waiting window [17] or a short waiting window [11, 30], e.g., 30 minutes, to mitigate the influence of data freshness issue. They apply various sample duplication mechanisms, which duplicate fake negatives with a corrected label after they receive conversions, and ingest them into the training pipeline again. Those methods adopt importance sampling [3] to adjust the weights of samples, making fake negatives less influential. Such techniques significantly reduce the gap between the ground truth distribution and the observed biased distribution, and achieve the state-of-the-art outcome [5].

Despite the success achieved by the current approaches, they still suffer from their short and limited waiting time. A short waiting window recognizes most samples with conversions as fake negatives initially and duplicates them with a positive label, resulting in massive samples with identical features but different labels in the data pipeline. Those confusing samples would disturb the estimation of the importance sampling [5], leading to sub-optimal

performance. Moreover, restricting the waiting windows to a quite short range cannot adapt to demands from real scenarios, considering the significant difference in delayed time windows among individual samples. Thus, a waiting window free from concerns about data freshness is expected.

In this paper, we propose an approach named DFSN (short for Delay Feedback modeling via neural Satellite Networks) for online CVR prediction. Opposite to previous methods, we assign a long waiting window for our main model to cover most of conversions and greatly reduce fake negatives. A duplication mechanism is applied to copy the remaining delayed positives and further mitigate the gap between our training data and the ground-truth data. Since one model could not simultaneously acquire data freshness and label accuracy, we devise two kinds of satellite models to learn from the latest data, and utilize online transfer learning skills, embedding transfer and strategy aggregation, to sufficiently exploit their knowledge. With information from satellites, our main model can deal with the issue of data freshness, achieving better performance than previous methods. Our method can be easily integrated with existing models, loosening the chain that jails the waiting windows and providing them with adaptivity and robustness.

The main contributions are summarized as follows:

- We emphasize the significance of an adaptive waiting window in online CVR training. This is essential for eliminating fake negatives and accustoming to various data distributions.
- We propose DFSN, which addresses the issue of data freshness by utilizing properly constructed satellites and transfer learning techniques. We provide existing methods with the adaptivity to adjust their waiting window for better performance.
- We conduct extensive experiments on two real-world CVR prediction datasets and demonstrate the state-of-the-art performance of our method.

The remainder of this paper is organized as follows. Section 2 surveys the related research in the literature. Section 3 introduces the preliminaries of this paper. Section 4 details the proposed DFSN framework, and Section 5 illustrates the experimental results. Section 6 concludes the paper.

2 RELATED WORK

2.1 Delayed Feedback Models

Online advertising places great emphasis on conversion rate (CVR) prediction [6, 18]. In-depth research has been done on conversion modeling in the literature. Chapelle [4] proposed DFM, two jointly-trained models for predicting the CVR and the delay time respectively under the assumption of an exponential distribution of delay. However, an exponential distribution is not always feasible for delayed feedback. In order to estimate the time delay without any assumptions about parametric distributions, Yoshikawa and Imai [33] developed DFM into a non-parametric delayed feedback model (NoDeF). In recent studies, conversions can be divided into various bins based on their delay time and measured separately [7, 25, 28]. Li et al. [19] built FTP, a multi-task prediction framework where each task tracks the feedback pattern over a range of time periods. Their method optimizes the aggregation strategy by an ideal CVR prediction model trained by historical data with complete feedback.

Table 1: Statistics of Criteo and Tencent dataset.

Dataset	# features	# conversions	# Samples	# average CVR	# log period	# attribution period
Criteo Dataset	17	3,619,793	15,898,863	0.2277	60 days	30 days
Tencent Dataset	19	624,411	22,601,402	0.0276	9 days	5 days

Those discussed methods with multi-task design treat various sub-models equally, our DFSN, by contrast, gives prominence to long wait time models.

2.2 Unbiased Estimation Models

The conversion delay makes it difficult for models to learn directly from real data distribution, but importance sampling [3, 11] helps bridge the gap. Existing techniques train on observed biased data with a well-designed importance weighting function to produce an unbiased estimate of conversions. To name some, Ktena et al. [17] proposed fake negative weighted (FNW) and fake negative calibration (FNC) methods, which initially label all samples as negative and duplicate samples with a positive label once they receive conversion feedback. Yasui et al. [32] developed a feedback shift importance weighting approach (FSIW) in which the model waits a certain period of time for the actual conversion to take place. However, as FSIW does not support data rectification, delayed positive samples cannot be effectively exploited. ES-DFM [30] maintains a waiting window and only duplicates positive samples that have no conversion in the window to lessen the prevalence of fake negatives. Chen et al. [5] separated observed samples into four types with adjusted importance weights and demonstrated a consistent improvement in performance across above models. The aforementioned methods also notice the challenge of data freshness. They mainly guarantee data freshness by using no waiting window or a very short one, while we propose a novel way to handle this issue.

2.3 Delay Bandits

The bandit learning method, is one alternative approach for solving the delayed feedback problem [15, 23, 27], which optimizes the model to minimize the cumulative regret throughout sequential decisions. Vernade et al. [26] proposed a stochastic multi-armed bandit model but fixed the delay distribution. This constraint is relaxed in the later work [27] by introducing a hyperparameter to control valid conversions. Wang et al. [29] computed the delay-corrected feedback by the Expectation-Maximization method to select best creatives based on a Thompson Sampling Bandit [24]. These bandit models aim to achieve the lowest regret whose objective is different from other CVR prediction methods.

3 PRELIMINARY

3.1 Data Pipeline

In this work, we focus on the online CVR prediction problem with delayed feedback. At time τ , the ground-truth dataset can be formulated as:

$$\mathcal{D}_\tau = \{(c_i, v_i, x_i, \hat{y}_i)\}_{i=1}^{N_\tau}, \quad (1)$$

where a sample has c_i the timestamp when it is logged, e.g. clicked, v_i the timestamp when a conversion action happens, x_i the input

Table 2: A summary of major notations used in the paper.

Notation	Explanation
c_i, v_i	the logged time and the conversion time
τ, d_{max}, d	the current time, the maximum attribution time and the waiting window
$x_i, \hat{y}_i, y_i$	the input feature, the ground-truth label and the observed label
$\mathcal{D}_\tau, \mathcal{D}_{\tau,d}, \mathcal{D}_{\tau,d}^+$	the ground-truth dataset, the observed dataset and the adjusted dataset
Φ, e	the embedding layer and the embedding

feature, and $\hat{y}_i \in (0, 1)$ the ground-truth label indicating whether a conversion takes place in the end. $\mathcal{D}_\tau$ contains all samples that are logged before time τ . Notice that the sample never has a conversion would be given $v_i \equiv \infty$, and only the conversion happens in the maximum attribution time d_{max} would be treated as valid conversion. Thus we can define the label $\hat{y}_i$ as:

$$\hat{y}_i = \begin{cases} 1, & v_i \leq c_i + d_{max}, \\ 0, & v_i > c_i + d_{max}. \end{cases} \quad (2)$$

The goal of the online CVR prediction problem is to continuously learn a function f with parameter θ that minimizes the following ideal loss:

$$\mathcal{L}_{ideal} = \sum_{(x_i, \hat{y}_i) \in \mathcal{D}_\tau} \ell(\hat{y}_i, f_\theta(x_i)), \quad (3)$$

where ℓ denotes the classification loss. However, we are unable to directly access the full ground-truth dataset because of the delayed feedback. Models have to wait for a maximum attribution time d_{max} to see the real label of a sample after it is logged, which can be intolerable in online services. The majority of models choose to wait for a proper time window $d < d_{max}$ to determine the label of each logged sample. We can define their observed training dataset:

$$\begin{aligned} \mathcal{D}_{\tau,d} &= \{(c_i, v_i, x_i, y_i)\}_{i=1}^{N_\tau} | c_i < \tau - d \\ &= \{(c_i, v_i, x_i, y_i)\}_{i=1}^{N_{\tau-d}}. \end{aligned} \quad (4)$$

$\mathcal{D}_{\tau,d}$ limits the training data to a subset logged before $\tau - d$ to make sure label y of every sample is always available at time τ , which would be:

$$y_i = \begin{cases} 1, & v_i \leq c_i + d, \\ 0, & v_i > c_i + d. \end{cases} \quad (5)$$

To combat the issue of fake negatives, recent methods use sample duplication mechanisms to clone mislabeled samples with a positive label after they receive conversion feedback and ingest them into training data again. Those duplicated samples for models with a short waiting window length d can be defined as:

$$\mathcal{P}_{\tau,d} = \mathcal{D}_\tau | c_i + d < v_i \leq \tau, \quad (6)$$

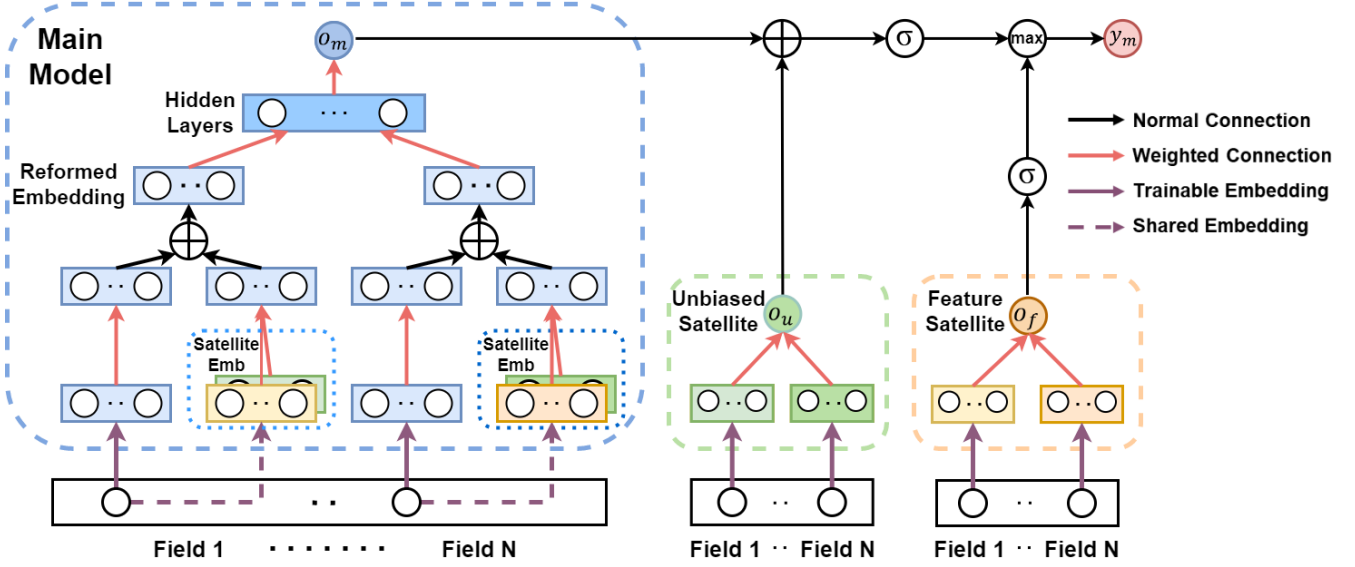


Figure 2: An illustration of DFSN, including the main model, the unbiased satellite model, and the feature satellite model. The main model utilizes embeddings and outputs of satellites to perform CVR prediction.

which contains all positive samples that had conversion before τ but were regarded as negative before. Every sample in $P_{\tau,d}$ would be correctly labeled as $y_i \equiv 1$. $P_{\tau,d}$ is available for training at time τ . So the adjusted training dataset would be:

$$\mathcal{D}_{\tau,d}^+ = \mathcal{D}_{\tau,d} \cup \mathcal{P}_{\tau,d}. \quad (7)$$

3.2 Importance Sampling

Recall that ES-DFM [30] with a short waiting window d is trained on the adjusted dataset $\mathcal{D}_{\tau,d}^+$. FNW/FNC [17] sets all samples as negatives at first and has no waiting window, its training dataset could be viewed as $\mathcal{D}_{\tau,0}^+$. These datasets with duplication samples have numerous fake negatives compared to the ground truth dataset. However, by using importance sampling, models training on biased datasets are able to approximate the distribution of ground-truth data. Models recast the loss function and set weights for samples to achieve this. Suppose we use binary cross-entropy loss, the weighted loss function for ES-DFM and FNW can be expressed as follows:

$$\mathcal{L}_{\text{ESDFM}} = - \sum_{(x_i, y_i) \in \mathcal{D}_{\tau,d}^+} y_i \{1 + p_{dp}(x_i)\} \log(f_{\theta}(x_i)) \quad (8)$$

$$+ (1 - y_i) \{(1 + p_{dp}(x_i))p_{rn}(x_i)\} \log(1 - f_{\theta}(x_i)),$$

$$\mathcal{L}_{\text{FNW}} = - \sum_{(x_i, y_i) \in \mathcal{D}_{\tau,0}^+} y_i \{1 + f_{\theta}(x_i)\} \log(f_{\theta}(x_i)) + \quad (9)$$

$$(1 - y_i) \{(1 + f_{\theta}(x_i))(1 - f_{\theta}(x_i))\} \log(1 - f_{\theta}(x_i)).$$

$\{\cdot\}$ means the stop gradient operation and terms inside are the weights of positive and negative samples. FNW uses its own prediction to calculate the weight, while ES-DFM trains auxiliary models to predict $p_{dp}(x)$, the probability of being a fake negative that has

conversion outside the waiting window, and $p_{rn}(x)$, the probability that a sample is a real negative and receives no valid conversion.

4 METHODOLOGY

In this section, we detail our proposed method, i.e. DFSN. First, we introduce the models in DFSN, including a main model that waits long enough to capture most of delayed feedback, and two kinds of auxiliary satellites training on the latest data. Then we illustrate the two techniques, namely embedding transfer and strategy aggregation, that utilize satellites to assist the main model in overcoming the issue of data freshness. At last, we give a brief discussion about the processing efficiency of DFSN. Figure 2 illustrates how DFSN cooperates with satellite models.

4.1 Satellites with Diverse Perspectives

Recall that our DFSN method employs a main model M_m with a lengthy waiting window d_m . The waiting window d_m is large enough to initially capture the majority of valid conversions. To further reach the late conversions that come out beyond the waiting window, the main model follows the existing duplication mechanism [17, 30] to copy those missed positives. The main model M_m is training on dataset $\mathcal{D}_{\tau,d_m}^+$. Two key divergences between the training dataset and the ground-truth dataset are as follows: Firstly, the training dataset $\mathcal{D}_{\tau,d_m}^+$ contains fake negatives as a result of delayed feedback, which has been significantly reduced by adopting the lengthy waiting window. However, the training dataset only includes examples that have been observed for a sufficient amount of time d_m , leaving out newly logged samples. The efficacy of the main model could be severely hampered by its inability to quickly identify the most recent trend.

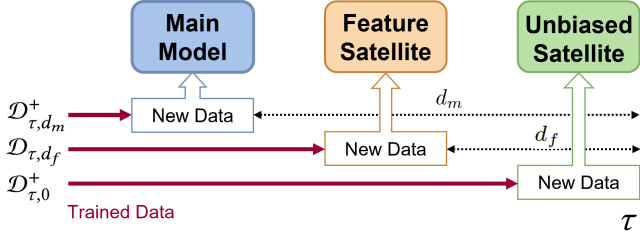


Figure 3: Data pipeline of DFSN, where the main model and two satellite models are training respectively with various waiting windows on streaming data. Wherein d_m is greater than d_f , and the unbiased satellite network has no waiting window.

To address the data freshness issue above, we present two kinds of auxiliary satellites that learn from the most current data: a *feature satellite* that has no duplication mechanism and maintains a short waiting window to see the latest ground-truth feature distribution, as well as an *unbiased satellite*, which applies importance sampling [3] to approach an unbiased estimation. Note that the number of satellites could be varied, and so is the importance sampling strategy. But for ease of description, we only use one model for each kind of satellite, and build the unbiased satellite model based on the fake negative weighted method [17], as it is the simplest. Hereafter we denote the feature satellite model as M_f , and it uses a short waiting window d_f and gets trained on $\mathcal{D}_{\tau, d_f}^+$. The unbiased satellite model is denoted by M_u , which has no waiting window but a duplication mechanism and uses $\mathcal{D}_{\tau, 0}^+$ to train. Figure 3 illustrates how the models in DFSN are trained on streaming data. With the increase of time τ , three models continuously receive new samples from their data pipelines and update parameters accordingly.

Notably, the training of two satellites may degrade the time efficiency of serving process. Although existing methods have proved that methods with similar scale and auxiliary models can be deployed online [5, 11, 30], it is beneficial to adopt further optimization. In the following sections, we come up with techniques that improve the performance and efficiency of models, e.g., the removal and freeze of layers.

4.2 Embedding Transfer

We now present the embedding transfer technique of DFSN, which can investigate input features from a fresh viewpoint by integrating the embedding layers of satellite models. We suppose the raw input x of samples contains N various features:

$$x = [x^1, x^2, \dots, x^N]. \quad (10)$$

As the input includes many sparse features, e.g. ID, all features are transformed into embeddings and concatenated (or using more complex combination methods). The embedding layer of the main model is:

$$\Phi_m = [\phi_m^1, \phi_m^2, \dots, \phi_m^N], \quad (11)$$

where ϕ_m^i denotes the transformation for feature x^i . The embedding layer of satellite models can be similarly defined as Φ_f and Φ_u . The concatenated embedding e_m for following dense layers can be

written as:

$$\begin{aligned} e_m &= [\phi_m^1(x^1), \phi_m^2(x^2), \dots, \phi_m^N(x^N)] \\ &= [e_m^1, e_m^2, \dots, e_m^N]. \end{aligned} \quad (12)$$

Since transfer learning techniques have achieved great success in machine learning fields, we adopt similar concepts to use the embedding layers Φ_f and Φ_u trained on the most recent data to generate embeddings for samples in $\mathcal{D}_{\tau, d_m}^+$ and reconstruct them with e_m . The reformed embedding $\hat{e}_m$ is:

$$\hat{e}_m^i = w_m^i e_m^i + w_f^i \phi_f^i(x^i) + w_u^i \phi_u^i(x^i), \quad (13)$$

$$\hat{e}_m = [\hat{e}_m^1, \hat{e}_m^2, \dots, \hat{e}_m^N], \quad (14)$$

where w_m , w_f and w_u correspond to the trainable weights of linear transformation. $\hat{e}_m$ instead of e_m would be fed into dense layers of the main model. Additionally, to lower the scale of parameters, the linear transformations are applied separately to each feature embedding e_m^i before concatenating. Moreover, as information of training data would be distributed across all model layers, we do not use hidden layers in satellite models, maximizing the knowledge retained at their embedding layers.

4.3 Strategy Aggregation

Despite embedding layers already giving the main model a unique outlook on the data, the sole output layer of satellite models contains important information as well. To fully exploit models M_f and M_u , we further obtain their output logits o_f and o_u for data (x, y) in $\mathcal{D}_{\tau, d_m}^+$. The outputs can then be aggregated with o_m from the main model in a straightforward but effective way:

$$y_m = \max(\sigma(o_m + o_u), \sigma(o_f)) = \max(p_m, \sigma(o_f)), \quad (15)$$

where σ is the sigmoid activation function. y_m is used as the final prediction, while p_m , the sum of o_m and o_u , is used to calculate the loss on the sample:

$$\mathcal{L}_m = -y \log(p_m) - (1 - y) \log(1 - p_m), \quad (16)$$

where y is the label of the sample. The feature satellite model M_f has neither duplication mechanisms nor importance sampling, would miss massive positives and underestimate the conversion probability. We also considered ensemble methods to aggregate the output of satellites. However, even with low weight, the output o_f still causes negative bias on predictions. Since a fairly high prediction from M_f may imply that many immediate conversions occur, we just take o_f as a lower bound, to improve the robustness of DFSN.

Notably, during the training of the main model, the layers from satellite models would be frozen and unchangeable. We use p_m instead of y_m to calculate the loss, guaranteeing that the main model could utilize every loss to update itself.

4.4 Processing Efficiency

For online applications on commercial platforms, we provide further discussions about the processing speed of the DFSN model from two aspects, the training process and the serving process.

During the online training process, delayed feedback models with importance sampling would wait for the output of auxiliary models to calculate the weights of samples [11, 30]. Similarly, DFSN

waits for the training process of satellites to obtain the latest embeddings from them. Generally, the common training speed of online models in industrial systems has reached minute-level in a distributed context. It only takes several minutes from the occurrence of samples to the end of satellite training. To fairly simulate the processing delay of satellites, we assume such a processing delay is 1 hour. It means the main model could only get the satellites updated 1 hour ago and without information about the current hour. By this assumption, the main model and satellites could update parallelly and synchronously, eliminating the extra waiting time. In the experiments, DFSN is evaluated with such limitations. The results show even with the delay, DFSN reaches a similar performance. The reason is that the satellites still contain quite recent information compared with the long waiting window of the main model. The contribution of satellites to freshness could be robust against such delay.

The other aspect related to efficiency is the online serving process. Following previous work with a multi-head architecture [19], the main model and satellites in DFSN are designed to generate output parallelly to aggregate the final prediction in Eq.(15). As satellites have simpler model structures, their calculation and prediction would be commonly faster than the main model. The efficiency of the serving process would not be disturbed by parallel satellites.

5 EXPERIMENTS

In order to validate our proposed model, experiments are conducted on two representative public advertising datasets. In this section, we first provide an overview of the design and implementation of experiments, and then demonstrate the significance of DFSN by responding to the following research questions:

- **RQ1:** How does DFSN perform in terms of online conversion rate prediction when compared to the state-of-the-art models?
- **RQ2:** How does DFSN address the issue of data freshness and enhance current methods?
- **RQ3:** How do satellites and components of DFSN affect its performance respectively?
- **RQ4:** How does the processing delay of satellites affect the performance of DFSN?

5.1 Datasets

The statistics of the two datasets are given in Table 1. We follow the original maximal attribution window setting of datasets in experiments, and the distribution of conversions is shown in Figure 1.

5.1.1 Criteo Conversion Logs. Criteo¹ is a widely used dataset for CVR prediction task [4]. It contains 60 days data, with a 30 days attribution window d_{max} . The average conversion rate of Criteo is 22.8%, representing a general scenario with relatively abundant feedback.

5.1.2 Tencent Advertising Algorithm Competition 2017. Tencent dataset² includes 14 days of click and conversion logs with a 5 days attribution window d_{max} . Data of last 5 days are excluded to fulfill the attribution window. The dataset contains 22 millions samples

with an average conversion rate of 2.76%, representing the real industrial scenarios with massive data but sparse feedback.

5.2 Experimental Settings

5.2.1 Online Simulation. The online streaming process is a scenario that models keep predicting CVR of new samples and then training on them. To imitate this, we separate the dataset into pieces by hour. Every piece involves samples that were logged during that hour, as well as information about observed conversion feedback. Models would be trained using available information from the present piece, and then evaluated on data from the next hour if needed. After that, we continue to the following hour and repeat this. To guarantee a fair comparison, the evaluation begins after a complete attribution period, e.g., 30 days in Criteo. Baseline methods with importance sampling [5, 30] are allowed to use all the data before evaluation to complete pre-training as they need.

5.2.2 Evaluation Metrics. We adopted three distinctive evaluation metrics to evaluate the model performance. The first is the area under the ROC curve (AUC), which denotes the ability to recognize and rank positive samples and negatives. The second is negative log likelihood (NLL), which is widely used and sensitive to the absolute value of the predictions [4]. Additionally, we introduce a distinct metric **Bias** that highly relates to industrial application and evaluates the global absolute values of predictions:

$$Bias_Z = \frac{1}{|\mathcal{D}|} \sum_{j=1}^{|Z|} N_{Z_j} \frac{|\sum_{z_i=Z_j, i \in \mathcal{D}} (y_i - p_i)|}{\sum_{z_i=Z_j, i \in \mathcal{D}} y_i}, \quad (17)$$

where $\mathcal{D}$ is the test dataset, Z is a selected field with discrete values, and samples with the same Z_j would be aggregated together. N_{Z_j} defines the number of samples with Z_j . y_i and p_i are the true label and model prediction value. Compared with NLL that measures instance-level accuracy, this metric evaluates the bias of total prediction values on various groups of samples. This metric matters because industries would require to estimate the total conversion for a certain class of advertisement [21]. Here we choose the **Campaign-ID** feature as Z . Both datasets have this feature.

Moreover, the purpose of delayed feedback models is to achieve similar performance with the ideal Oracle model that gets trained on data without delay. To compare with the ideal model, we define the relative improvement of AUC as:

$$RI - AUC_{DFSN} = \frac{AUC_{DFSN} - AUC_{Pretrain}}{AUC_{Oracle} - AUC_{Pretrain}}, \quad (18)$$

where **Oracle** is the model trained on the ideal dataset without delay, and **Pretrain** represents the naive baseline for importance sampling models [5, 30]. RI-NLL and RI-Bias can be defined similarly. Following previous works [11, 30, 31], relative metrics are used in the main experiments to illustrate the performance of models.

5.2.3 Compared Methods.

- **Oracle:** An ideal model training on dataset $\mathcal{D}_\tau$ with the ground-truth label, representing the upper bound.
- **Pretrain:** A model only gets pre-trained on data before evaluation and has no streaming training process, representing the lower bound.
- **Vanilla:** A model training with a finetuned waiting window.

¹<https://labs.criteo.com/2013/12/conversion-logs-dataset/>

²<https://algo.qq.com/?lang=en>

Table 3: Performance comparisons of the proposed model with baseline models on AUC, NLL, and Bias metrics. The Pretrain method and the Oracle method respectively correspond to the lower bound 0% and the upper bound 100%, their absolute performance is in parentheses. The best value in one column is displayed by the bold value, and the second is indicated by the underlined value. * indicates statistical significance improvement compared to the best baseline measured by t-test at p-value of 0.05.

Datasets	Criteo Dataset			Tencent Dataset		
Metrics	AUC	NLL	Bias	AUC	NLL	Bias
Pretrain	0.0%(0.833)	0.0%(0.407)	0.0%(0.368)	0.0%(0.775)	0.0%(0.108)	0.0%(0.737)
Vanilla	27.5%	33.9%	51.7%	70.0%	60.3%	82.0%
FNW [17]	46.0%	49.2%	<u>89.6%</u>	74.3%	70.7%	86.4%
FNC [17]	44.9%	22.8%	88.6%	73.0%	67.2%	85.6%
ES-DFM [30]	64.6%	74.0%	88.3%	70.0%	65.5%	80.7%
DEFUSE [5]	66.9%	-19.3%	-99.8%	70.2%	-70.7%	-65.7%
FTP [19]	59.5%	50.8%	64.9%	79.9%	72.4%	85.0%
DFSN- α	73.0%*	78.3%*	85.5%	85.3%*	82.8%*	<u>88.4%*</u>
DFSN- β	<u>70.2%*</u>	<u>75.2%*</u>	90.7%*	<u>84.7%*</u>	<u>79.3%*</u>	89.9%*
Oracle	100%(0.851)	100%(0.382)	100%(0.075)	100%(0.818)	100%(0.102)	100%(0.065)

- **FNW [17]**: A model training with a duplication mechanism using the fake negative weighted loss.
- **FNC [17]**: A model training with a duplication mechanism with the fake negative calibration.
- **ES-DFM [30]**: A model training with a duplication mechanism using the ES-DFM loss.
- **DEFUSE [5]**: A model training with a duplication mechanism using the DEFUSE loss.
- **FTP [19]**: A model training with a multi-task learning and aggregation mechanism.
- **DFSN- α** : The proposed model training with a finetuned waiting window for best AUC.
- **DFSN- β** : The proposed model training with a finetuned waiting window for lowest Bias.

We choose DEFUSE instead of Bi-DEFUSE [5] because the former achieved much better results on the dataset with a long attribution window in the original paper. We also tried DEFER [11], but found it performed below Vanilla on Tencent dataset so we did not select it for comparison. As DFSN has an adaptive waiting window, we especially propose two models finetuned for best AUC and lowest Bias respectively.

5.2.4 Parameter Settings. We implement the DFSN in Tensorflow. We use a DNN model with a fixed hidden size [128,128] as the base model for all the methods. Each hidden layer is followed by the Leaky ReLU activation function [20]. The features are initially transformed into embedding vectors and then fed to fully-connected layers above. L2 regularization is set to 10^{-6} on Criteo Dataset and 10^{-7} on Tencent Dataset. The models are updated by the Adam optimizer [16]. For a fair comparison, we apply the grid search strategy to tune the best learning rate among {0.0001, 0.0005, 0.001}, and carefully tune the short waiting window for previous models in accordance with the original papers [5, 30].

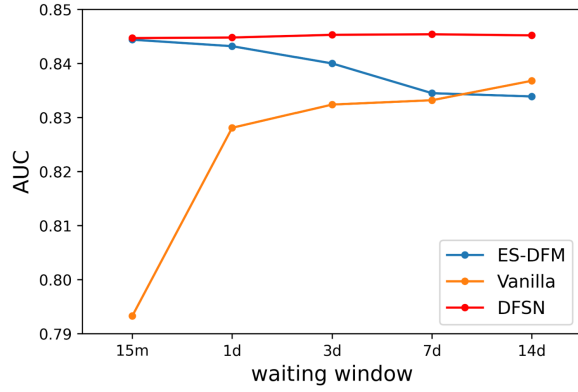
5.3 Main Experiments (RQ1)

We execute 5 random runs on Criteo and Tencent to illustrate the overall performance of DFSN and baselines. The averaged results with significance test are reported in Table 3. We have the following findings upon their performance comparison:

- Our DFSN method yields remarkable improvements over the baselines on both datasets. In particular, DFSN- α outperforms the strongest baselines w.r.t. RI-AUC by 6.1%, 5.4% on Criteo and Tencent dataset, respectively. It is notable that DFSN- β already outperforms all baselines at all metrics, while DFSN- α reaches further improvement at AUC and NLL metrics. As indicated by previous research [5, 36], even 0.1% improvement in offline AUC can result in a sizable increase in online advertising. Our strategy is successful because it addresses the issue of outdated data and provides a new, adaptive waiting window to tackle various challenges and demands. DFSN- β tunes the waiting window to meet the requirement of Bias and obtains significant RI-Bias over 80% on Criteo.
- Conversions in Tencent dataset follow a long tail distribution that more than 90% conversions happen in the first day. DFSN and FTP contain long waiting windows to directly model the delayed feedback in the first day without creating many fake negatives, and considerably outperform other state-of-the-art methods. However, ES-DFM and DEFUSE only get mediocre performance on Tencent dataset. One explanation is that, as ES-DFM and DEFUSE cannot use importance sampling until their importance weight models are ready, they utilize a pre-training model instead, which fails to learn the conversion distribution well. To test the upper bound of models, we allow ES-DFM, DEFUSE to train their auxiliary models in advance and use importance sampling during the whole training, which would boost performance but is impractical. For a comprehensive comparison, We also evaluate DFSN applying importance sampling like ES-DFM. The

Table 4: Results for models with enhanced importance sampling on Tencent Dataset.

Metrics	AUC ($\uparrow$)	NLL ($\downarrow$)	Bias ($\downarrow$)
ES-DFM+IS	0.8072	0.1041	0.1544
ES-DFM	0.8053(-4.1%)	0.1044(-5.1%)	0.1944(-5.9%)
DEFUSE+IS	0.8086	0.1114	0.9878
DEFUSE	0.8054(-7.5%)	0.1123(-15.5%)	1.1797(-28.5%)
DFSN+IS	0.8122	0.1035	0.1424
DFSN	0.8120(-0.6%)	0.1036(-1.7%)	0.1427(-0.1%)

**Figure 4: The effect of different waiting windows.**

results are given in Table 4, both ES-DFM and DEFUSE suffer a heavy decrease at all metrics from their weak pre-training model, while DFSN does not. That is because our satellites make the model robust to datasets with various distributions. Even without the trick above, our model still outperforms those enhanced baselines by a great margin, indicating the benefit of the adaptive waiting window can reach the best wait time.

- Vanilla performs better than expected if its waiting window is properly tuned. This suggests that models with a long waiting window do have potential [19]. The higher score of FNW and FNC is indicative of the significance of the duplication mechanism and importance sampling. Since ES-DFM maintains a short waiting window to decrease the amount of fake negatives, it outperforms earlier models significantly. DEFUSE builds a hidden model to precisely compute importance weight and reaches best AUC among baselines on Criteo. However, sophisticated terms are contained in its calculation, resulting in a relatively high NLL and Bias.

5.4 Data Freshness (RQ2)

To demonstrate how DFSN tackles the issue of data freshness, we compare the AUC of Vanilla, ESDFM, and DFSN with various waiting windows on Criteo dataset. For straightforward comparisons here, DFSN uses ES-DFM as its main model. The results are shown in Figure 4, here are some observations:

Table 5: Ablation study for two kinds of satellites.

Metrics	AUC ($\uparrow$)	NLL ($\downarrow$)	Bias ($\downarrow$)
DFSN	0.8458	0.3870	0.1177
DFSN w/o U	0.8437(-26.5%)	0.3894(-25.3%)	0.1544(-12.6%)
DFSN w/o F	0.8436(-27.3%)	0.3894(-25.3%)	0.1122(+1.8%)
DFSN w/o U, F	0.8386(-66.0%)	0.3958(-63.4%)	0.1882(-24.1%)

- Vanilla with a short waiting window, e.g., 15 minutes, results in poor AUC for missing massive positive samples. With the increase of the waiting window, Vanilla performs better as it could catch more delayed conversions, indicating the advantage of longer observation time.
- The AUC of ES-DFM reaches the top initially and then continues to decline. Although a long waiting window helps ES-DFM to reduce fake negatives, the price of losing data freshness could be unbearable.
- For models without access to the latest information, the harm of losing data freshness outweighs the benefit of long waiting windows. However, employing satellite models on ES-DFM gets over this issue and further improves AUC by eliminating most fake negatives. From our experimental results, the best waiting window for DFSN on Criteo dataset is around 10 days.

5.5 Ablation Study (RQ3)

To specifically validate the effectiveness of our design, we conduct ablation studies to evaluate the contributions of each component of DFSN on Criteo, respectively.

5.5.1 Unbiased Satellite and Feature Satellite. Two kinds of satellites with different data pipelines are employed in DFSN to access the latest information. To analyze their effect, we evaluate four formulations of DFSN:

- **DFSN:** The proposed DFSN with two satellite models.
- **DFSN w/o U:** DFSN without the unbiased satellite.
- **DFSN w/o F:** DFSN without the feature satellite.
- **DFSN w/o U, F:** DFSN without any satellite.

The results are shown in Table 5. Both satellite models bring apparent improvement to DFSN w/o U, F. DFSN performs much better than DFSN w/o U and DFSN w/o F, suggesting that the unbiased satellite and the feature satellite are complementary and enhance the main model with their distinctive perspectives. It also indicates that a satellite learning the latest feature distribution is of great importance. Besides, formulations without the unbiased satellite model result in higher Bias. We notice that DFSN w/o F has a lower Bias than DFSN, which suggests more satellite models may be harmful to performance on Bias. Due to this, even if we discovered that adding more satellite models could further enhance AUC, we opted to maintain the number of satellite models at two. However, it does not hinder the proposed DFSN from being a flexible framework that can be adjusted according to the different demands of users.

5.5.2 Embedding Transfer Strategies. As introduced in Section 4.2, we use linear transformation to reform the embeddings of models, transferring the knowledge of satellites to the main model. The reformed embedding contains information from three models. We

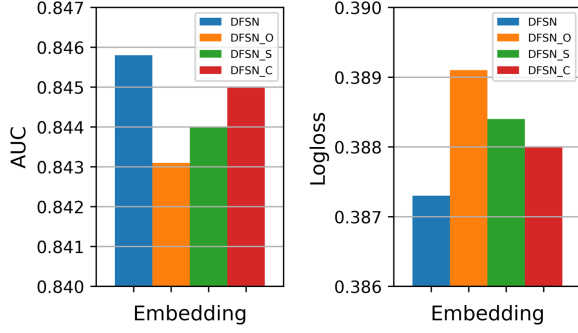


Figure 5: The performance of different embedding strategies.

Table 6: Ablation study for various aggregation components.

Metrics	AUC (↑)	NLL (↓)	Bias (↓)
DFS	0.8458	0.3870	0.1177
DFS w/o o_u	0.8450	0.3876	0.1212
DFS w/o o_f	0.8452	0.3874	0.1156
DFS w/o o_u, o_f	0.8445	0.3882	0.1242

further investigate the influence of following embedding strategies:

- **DFS**: The proposed DFSN with the reformed embedding.
- **DFS_O**: DFSN directly using the original embedding from the main model.
- **DFS_S**: DFSN only reforming the embeddings from two satellites without the embedding from the main model.
- **DFS_C**: DFSN using concatenation instead of linear transformation to reform embeddings.

The results are shown in Figure 5. The proposed DFSN consistently achieves better AUC and lower Logloss than DFSN_O and DFSN_S, suggesting that the removal of embeddings from any models is always detrimental. Embedding layers of three models get trained on various data pipelines and contain unique knowledge. Besides, since DFSN outperforms DFSN_C, the linear transformation could be more appropriate for embedding reformation than concatenation.

5.5.3 Aggregation Strategies. To further investigate the knowledge stored in the output layer of satellites, we verify the effect of the strategy aggregation shown as Eq. 15. We separately remove the output of models to evaluate their contributions:

- **DFS**: The proposed DFSN with the prediction given in Eq. 15.
- **DFS w/o o_u** : DFSN without the output of the unbiased satellite.
- **DFS w/o o_f** : DFSN without the output lower bound from the feature satellite.
- **DFS w/o o_u, o_f** : DFSN directly using the output of the main model without aggregation.

The results are given in Table 6. DFSN w/o o_u and DFSN w/o o_f both outperform DFSN w/o o_u, o_f , suggesting the benefit of aggregation. However, the improvement brought by aggregation is relatively modest since most of information could already be utilized by sharing embeddings. Notably, by taking the feature satellite

Table 7: Results for DFSN without satellites processing delay.

Metrics	AUC (↑)	NLL (↓)	Bias (↓)
DFS	0.8458	0.3870	0.1177
DFSN w/o delay	0.8458	0.3871	0.1175

output as a lower bound, the AUC of DFSN further increases, indicating the necessity of tracking latest feedback.

5.6 Processing Efficiency (RQ4)

Recall that in Section 4.4, we set the training time of satellites as 1 hour to simulate the training delay in online scenarios with massive samples. For comparison, we remove the limitation and allow the main model to get trained after the update of satellites, obtaining the latest embedding from satellites. According to the results in Table 7, DFSN without the processing delay still has a similar performance. The training delay in real scenarios has little influence on the freshness contribution of satellites, suggesting the robustness of our proposed model. Besides, we also evaluate the predicting time of the main model and satellites. Satellites with fewer layers only take 63% of time to calculate predictions compared with the main model, thus would not affect the serving efficiency.

5.7 Online Simulation

We finally conducted an online simulation test on real traffic data from online scenarios to evaluate the proposed DFSN model. More than 10 thousand features are included in each sample. The valid attribution window is 7 days. We set the waiting window of the main model as 5 days and limit the maximum processing time for satellites to 1 hour. By introducing the satellites, we observe 5.7% improvement in the CVR task. This aligns with previous results and demonstrates the effectiveness of DFSN in industrial systems.

6 CONCLUSION

In this paper, we concentrated on the online CVR prediction task and proposed the DFSN approach to deal with the issue of delayed feedback. By using satellite networks that extract knowledge from the latest data, we aim to address the problem of data freshness. We employed techniques inspired by transfer learning and considerably exploited the knowledge of satellite networks by collecting their embedding layers and outputs. We also provide a new perspective for existing models with short and constrained waiting windows. That is the waiting windows could be adjusted to integrate different demands. Extensive experiments demonstrate that the proposed method clearly outperforms previous state-of-the-arts.

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